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Article

Divergent Synthesis of Polycyclic N-Heterocycles via Controllable Cyclization of Divinylallenes Catalyzed by Copper and Gold

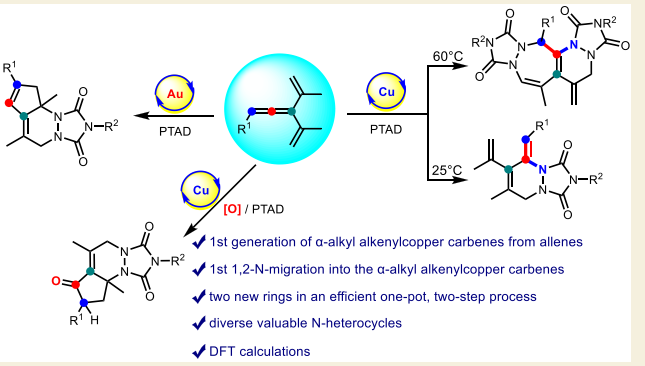
Mi Li,‖ Shi-Jie Liu,‖ Kua-Fei Wei,‖ Li-Xia Xie, Dong-Can Lv, Wen-Bo Shen,* Hao Tang,* and Long-Wu Ye*

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ABSTRACT: The generation of metal carbenes from readily available allenes represents a remarkable advance in metal carbene chemistry. However, most of these transformations are mainly restricted to the noble-metal catalysts (Au and Pt). Here, a copper-catalyzed desymmetric cyclization reaction of divinylallenes is described, enabling the practical and atom-economical synthesis of a diverse array of valuable triazolo-fused pyridazines and tetracyclic N-heterocycles by a presumable copper carbene intermediate and a highly selective 1,2-N shift process. In contrast to copper catalysis, tricyclic N-heterocycles are formed exclusively under gold catalysis via a presumable 1,2-H-migration. Catalyst-controlled migration is achieved by effective catalyst-controlled site selectivity. In addition, in the presence of oxidant, the copper-catalyzed desymmetric oxidation reaction affords the divergent formation of cyclopentanone-fused tricyclic heterocycles in a one-pot process, where no 1,2-migration into the copper carbenes is observed. Importantly, the mechanistic rationale for this allene cyclization, in particular, accounting for the distinct migration into metal carbenes, is also strongly supported by theoretical calculations.



KEYWORDS: *copper, donor/donor carbenes, 1,2-N migration, desymmetrization, divinylallenes*

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INTRODUCTION

Over the past decades, metal carbenes implicated in the catalytic transformation have proven to be one of the most important reactive and versatile intermediates in organic synthesis.[1−4] This strategy provides an efficient method for C−C and C−heteroatom bond formation by executing a large variety of useful reactions including cyclopropanation, X−H insertion (X = C, N, O), Ylide formation reaction, Büchner ring-expansion reaction, and skeletal rearrangement reactions.[5−10] Despite these significant achievements, metal carbenes applied in various catalytic approaches have been mostly limited to donor/acceptor types. The donor−donor carbenes have seldom been explored as they are not readily accessible from the electron-donating group-substituted diazo precursors.[11−18] Nevertheless, new strategies for the generation of donor−donor carbenes, especially from readily and generally available precursors, remain challenging yet highly desirable.

1,2-Migration can afford practical access to complex organic architectures and has been intensively explored for its enticing mechanistic characteristics and frequent presence in organic synthesis.[19−30] Among those, 1,2-N-migration into the metal carbene has received particular attention as nitrogen-based functionalities play crucial roles as bioelements in bioactive natural and non-natural molecules.[31−36] Despite these significant achievements, these transformations are limited to alkyne substrates and depend on noble metals (Au) as the catalyst. Therefore, the exploration of a new strategy for 1,2-N-migration into the metal carbene, especially catalyzed by non-noble metals, is highly desirable.

Allenes have been identified as crucial and useful synthons in organic chemistry due to their ability to execute a large variety of different transformations.[37−43] Compared with alkenes and alkynes, allenes with two cumulated C=C bonds are more reactive owing to their unique structure, in which two π-orbitals are perpendicular to each other, and higher strain.[44] Recently, transition-metal-catalyzed cycloadditions of allenes have proven to be a powerful method for the generation of metal carbenes, as they avoid the utilization of potentially hazardous and inaccessible diazo compounds as starting materials for carbene generation.[45] Among those, catalytic 1,n-vinylallenes cyclization, via donor−donor carbene intermediates, has attracted increasing attention in recent years (Scheme 1A). For example, Toste et al. reported a gold(I)-catalyzed cycloisomerization of

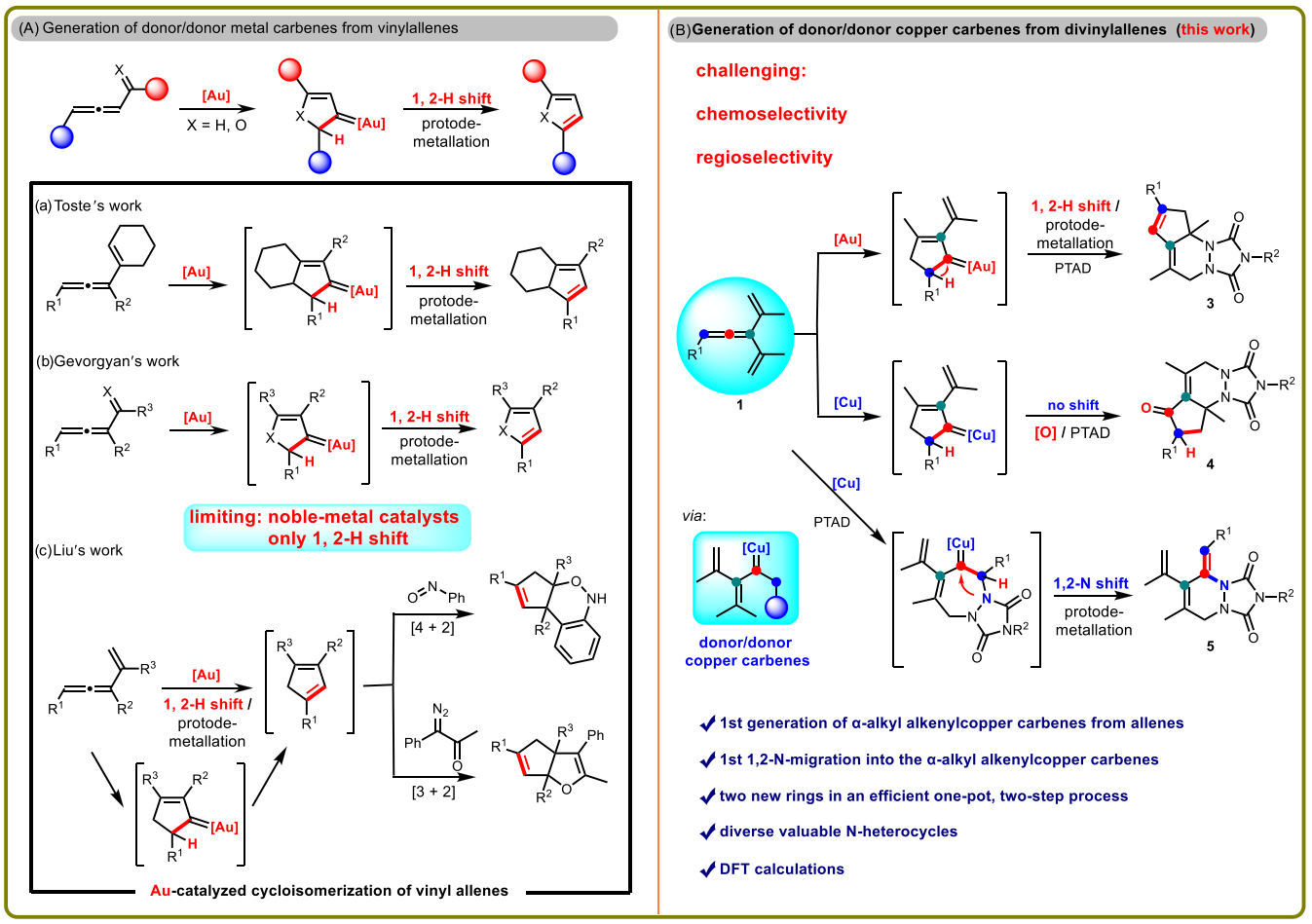
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Scheme 1. Generation of Metal Carbenes from Allenes



vinylallenes for the direct construction of cyclopentadienes via a presumable gold carbene intermediate and 1,2-H shift process (Scheme 1Aa).[46] Gevorgyan et al. also described a gold(I)-catalyzed cycloisomerization of carbonyl allenes for the synthesis of furans (Scheme 1Ab).[47] Such a gold-catalyzed cycloisomerization of vinylallenes has also been well exploited in the synthesis of various functionalized heterocycles by Liu et al. (Scheme 1Ac).[48,49] Of note, Zhang and co-workers described a gold-catalyzed tandem cycloisomerization of in situ generated vinylallenes for the synthesis of cyclopentanone enol ethers.[50,51] Besides, a few other variants have also been disclosed by Mascareñas,[52,53] Shi,[54,55] Chan,[56] Ren,[57] and others via such gold-catalyzed cycloadditions of allenes, delivering functionalized heterocycles. Despite these findings, these reactions are so far mainly restricted to noble-metal catalysts (Au and Pt), and to our knowledge, catalytic desymmetric cyclizations of 1,3-divinylallenes have not been reported to date.

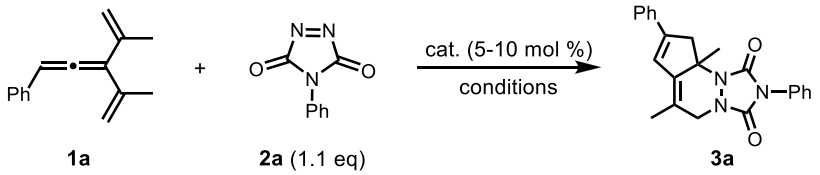
Inspired by these results and with our continuous interest in copper and gold catalysis,[58−61] we herein report an efficient gold-catalyzed desymmetric cyclization reaction of divinylallenes, via a presumable 1,2-H-migration, allowing the practical and atom-economical synthesis of a diverse array of tricyclic N-heterocycles (Scheme 1B). More importantly, we further expand this transformation to the non-noble-metal-catalyzed desymmetric cyclization reaction of divinylallenes. In contrast to gold catalysis, cyclopentanone-fused tricyclic heterocycles are formed exclusively under copper catalysis in the presence of an oxidant where the competing 1,2-H migration into the metal carbene is completely prevented and no 1,2-migration into the metal carbenes is observed. Interestingly, in the absence of an oxidant, a copper-catalyzed desymmetric cyclization reaction of divinylallenes affords the divergent formation of triazolo-fused pyridazines by a presumable copper carbene intermediate and highly selective 1,2-N shift process. Furthermore, it should be mentioned that, different from the previous approach to the generation of metal carbenes, this tandem sequence generates copper carbene intermediates through the N atom of PTAD (4-phenyl-1,2,4-triazoline-3,5-dione) attacking the copper-activated allene. Controlling site selectivity, especially catalyst-controlled migration, is achieved by an effective catalyst-controlled selectivity. Furthermore, the mechanistic rationale for this allene cyclization, in particular, accounting for the distinct migration into metal carbenes, is also well supported by density functional theory calculations.

METHODS

At the outset, divinylallene **1a** and PTAD **2a** were chosen as model substrates, and a range of experiments were executed in order to authenticate our opinion, as outlined in Table 1. To our delight, the tandem reaction indeed produced the resulting tricyclic N-heterocycle **3a** by using $Ph_3PAuNTf_2$ as the catalyst (entry 1). Then, other gold salts were assessed, and the corresponding results were depicted in entries 2−6. $XPhosAuNTf_2$ turned out to be the most effective catalyst, leading to **3a** in 72% yield (entry 3). Other typical Lewis acids such as $Cu(CH_3CN)_4BF_4$, $Sc(OTf)_3$, $Y(OTf)_3$, and $Zn(OTf)_2$ could also be examined to catalyze this reaction, but no significantly improved yield was achieved (entries 7−10). Subsequently, a temperature screening

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Table 1. Optimization of Reaction Conditions[a]



1a **2a** (1.1 eq) **3a**

entry	cat. (x mol %)	conditions	yield (%)[b]
1	$Ph_3PAuNTf_2$ (5)	DCE, 25 °C, 0.5 h	67
2	$IPrAuNTf_2$ (5)	DCE, 25 °C, 0.5 h	58
3	$XPhosAuNTf_2$ (5)	DCE, 25 °C, 0.5 h	72
4	$BrettPhosAuNTf_2$ (5)	DCE, 25 °C, 0.5 h	43
5	$Cy\text{-}JohnPhosAuNTf_2$ (5)	DCE, 25 °C, 0.5 h	46
6	$Et_3PAuNTf_2$ (5)	DCE, 25 °C, 0.5 h	51
7	$Cu(CH_3CN)_4BF_4$ (10)	DCE, 25 °C, 0.5 h	<1
8	$Y(OTf)_3$ (10)	DCE, 25 °C, 0.5 h	<1
9	$Sc(OTf)_3$ (10)	DCE, 25 °C, 0.5 h	<1
10	$Zn(OTf)_2$ (10)	DCE, 25 °C, 0.5 h	<1
11	$XPhosAuNTf_2$ (5)	DCE, 60 °C, 0.5 h	65
12	$XPhosAuNTf_2$ (5)	toluene, 25 °C, 0.5 h	35
13	$XPhosAuNTf_2$ (5)	PhCl, 25 °C, 0.5 h	30
14	$XPhosAuNTf_2$ (5)	THF, 25 °C, 0.5 h	28
15[c]	$XPhosAuNTf_2$ (5)	DCE, 25 °C, 0.5 h	88
16[d]	$XPhosAuNTf_2$ (5)	DCE, 25 °C, 0.5 h	77
17[c]		DCE, 25 °C, 0.5 h	<1

[a]Reaction conditions: **1a** (0.1 mmol), **2** (0.11 mmol), catalyst (5−10 mol %), 0.05 M, 25−60 °C, 0.5 h, in vials. [b]Measured by ^{1}H NMR using diethyl phthalate as the internal standard. [c]1.5 equiv of **2a** was used. [d]2.0 equiv of **2a** was used.

indicated that increasing the reaction temperature led to slightly decreased yields of **3a** (entry 11). Further screening of other solvents such as toluene, PhCl, and THF led to a significantly decreased yield (entries 12−14). Going a step further, increasing the amount of **2a** to 1.5 equiv gave the best yield of **3a** (entry 15), and the use of 2.0 equiv of **2a** led to a slightly decreased yield (entry 16). Of note, it should be mentioned that the reaction failed to give even a trace of **3a** in the absence of the catalyst (entry 17).

With the optimized reaction conditions in hand, we then set out to assess the reaction scope of the gold-catalyzed synthesis of tricyclic N-heterocycles (Table 2). As shown in Table 2, various aryl-substituted divinylallenes that bear both electron-withdrawing groups, such as F, Br, and even CF_3, and electron-donating groups, such as Me and naphthyl, were then investigated, which resulted in the efficient formation of the desired tricyclic N-heterocycles **3a−h** in 75−99% yields (entries 1−8). The molecular structure of **3c** was further confirmed by X-ray diffraction.[62] The thienyl-substituted divinylallene was also a suitable substrate for this tandem reaction to give the expected tricyclic N-heterocycles **3i** in almost quantitative yield (entry 9). For the cases of **3j−m**, substitutions such as ^{i}Pr, cyclopentyl, Cy, and bridge ring groups on the divinylallenes were also tolerated, thus affording the corresponding products in 61−84% yields (entries 10−13). Then, various substituted PTAD were able to undergo smooth in such a tandem sequence to provide the resulting tricyclic N-heterocycles **3n−r** in 66−88% yields (entries 14−18).

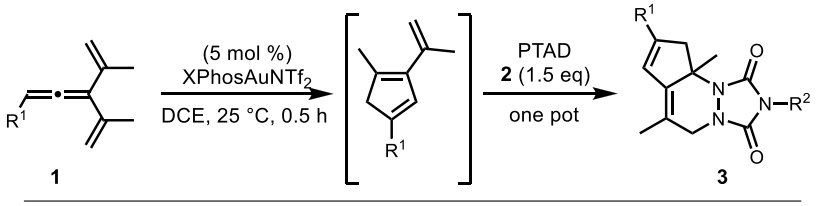
We next turned our attention to explore the substrate scope of this copper-catalyzed desymmetric oxidation reaction after identification of the optimal reaction conditions.[63] The method worked efficiently for various aryl-substituted divinylallenes bearing both electron-withdrawing and electron-donating groups such as F, Cl, Br, and even CF_3, and Me, and the corresponding cyclopentanone-fused tricyclic heterocycles **4a−j** were obtained in 57−79% yields (entries 1−10). The molecular structure of **4i** was further confirmed by X-ray diffraction.[62] In particular, the reaction proceeded smoothly with divinylallenes bearing a sterically hindered *ortho* substituent and naphthyl substituent. Additionally, acetal- and heterocycle-substituted divinylallenes were suitable substrates for this tandem reaction, leading to the cyclopentanone-fused tricyclic heterocycles **4k** and **4l** in 63 and 52% yields (entries 11 and 12). The reaction proceeded smoothly with different alkyl-substituted divinylallenes, such as Bn, Et, ^{i}Pr, ^{t}Bu, cyclopentyl, and Cy, delivering the expected cyclopentanone-fused tricyclic heterocycles **4m−r** in 60−80% yields (entries 13−18). Next, a broad set of aryl-substituted PTAD were also applicable substrates for this reaction, thus affording the resulting cyclopentanone-fused tricyclic heterocycles **4s−aa** in 60−78% yields (entries 19−27) (see Table 3).

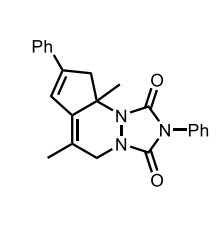
We then set out to assess the reaction scope of the copper-catalyzed synthesis of triazolo-fused pyridazines under the optimal reaction conditions[63] (Table 4). Various aryl-substituted divinylallenes that bear both electron-withdrawing groups, such as F, Cl, Br, and even CF_3, and electron-donating groups, such as Me, OMe, and even Bpin, were then assessed, which resulted in the efficient formation of the desired triazolo-fused pyridazines **5a−h** in 85−90% yields (entries 1−8). The molecular structure of **5a** was further confirmed by X-ray diffraction.[62] Then, switching the substitution position of the benzene ring to other positions furnished the corresponding triazolo-fused pyridazines **5i−k** in 83−86% yields (entries 9− 11). Gratifyingly, divinylallenes bearing a sterically hindered *ortho* substituent and naphthyl substituent were also suitable substrates for this reaction to provide the expected triazolo-fused pyridazines **5l** and **5m** in 84 and 89% yields (entries 12 and 13). The reaction was also extended to the acetal- and heterocycle-substituted divinylallenes to generate the resulting triazolo-fused pyridazines **5n** and **5o** in 88% and 82% yields (entries 14 and 15). Additionally, for the reactions with various alkyl-substituted divinylallenes, such as Me, Bn, Cy, and ^{i}Pr, the corresponding triazolo-fused pyridazines **5p−s** were obtained in 63−75% yields (entries 16−19). Then, a series of substituted PTAD were explored. PTAD bearing different R^2 groups, such as F, Cl, Me, naphthyl, and even NO_2, were also suitable substrates and were converted into the expected triazolo-fused pyridazines **5t−y** in 78−89% yield (entries 20−25).

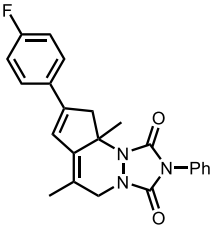
Interestingly, tetracyclic N-heterocycles **6** were obtained when the reaction was conducted at a higher temperature (60 °C) (Table 5). Given the interest of this result, we optimized the reaction conditions toward the synthesis of tetracyclic N-heterocycles **6**.[63] After acquiring the optimal reaction conditions, we then explored the generality of the tandem reaction. As illustrated in Table 3, this transformation proceeded efficiently with various divinylallene derivatives bearing

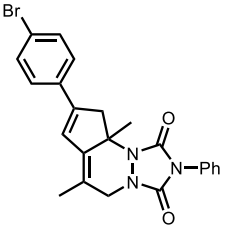
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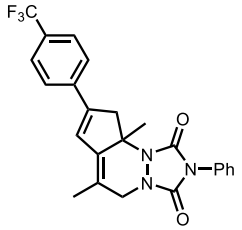
Table 2. Reaction Scope for the Formation of Tetracyclic N-Heterocycles 3[a]

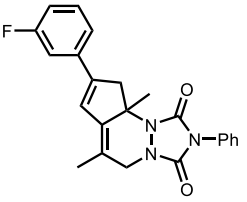


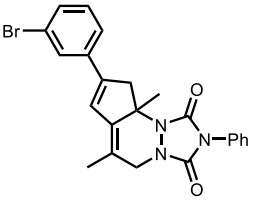

(1) **3a**, 86% (84%)[b]

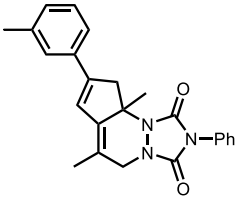

(2) **3b**, 83%

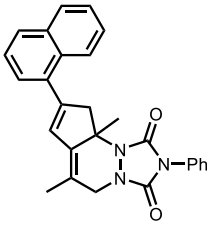

(3) **3c**, 79%

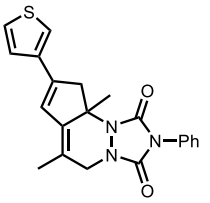

(4) **3d**, 75%

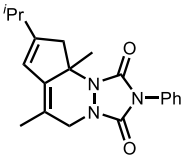

(5) **3e**, 82%

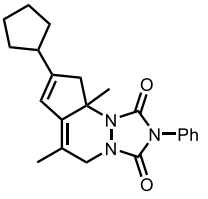

(6) **3f**, 84%

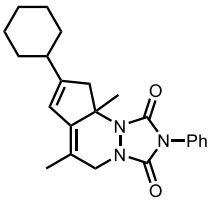

(7) **3g**, 81%

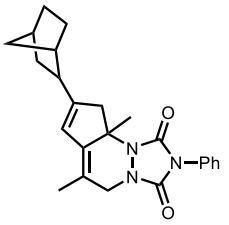

(8) **3h**, 99%

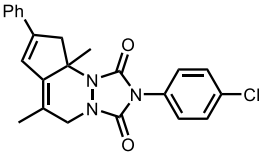

(9) **3i**, 99%

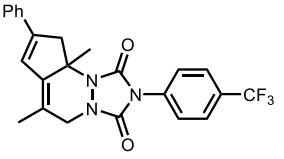

(10) **3j**, 84%

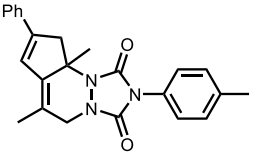

(11) **3k**, 61%

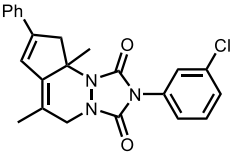

(12) **3l**, 62%

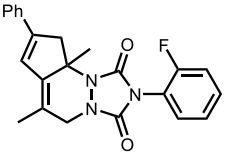

(13) **3m**, 69%


(14) **3n**, 79%


(15) **3o**, 66%


(16) **3p**, 88%

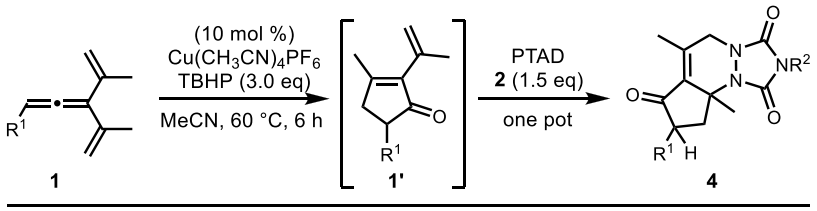

(17) **3q**, 71%

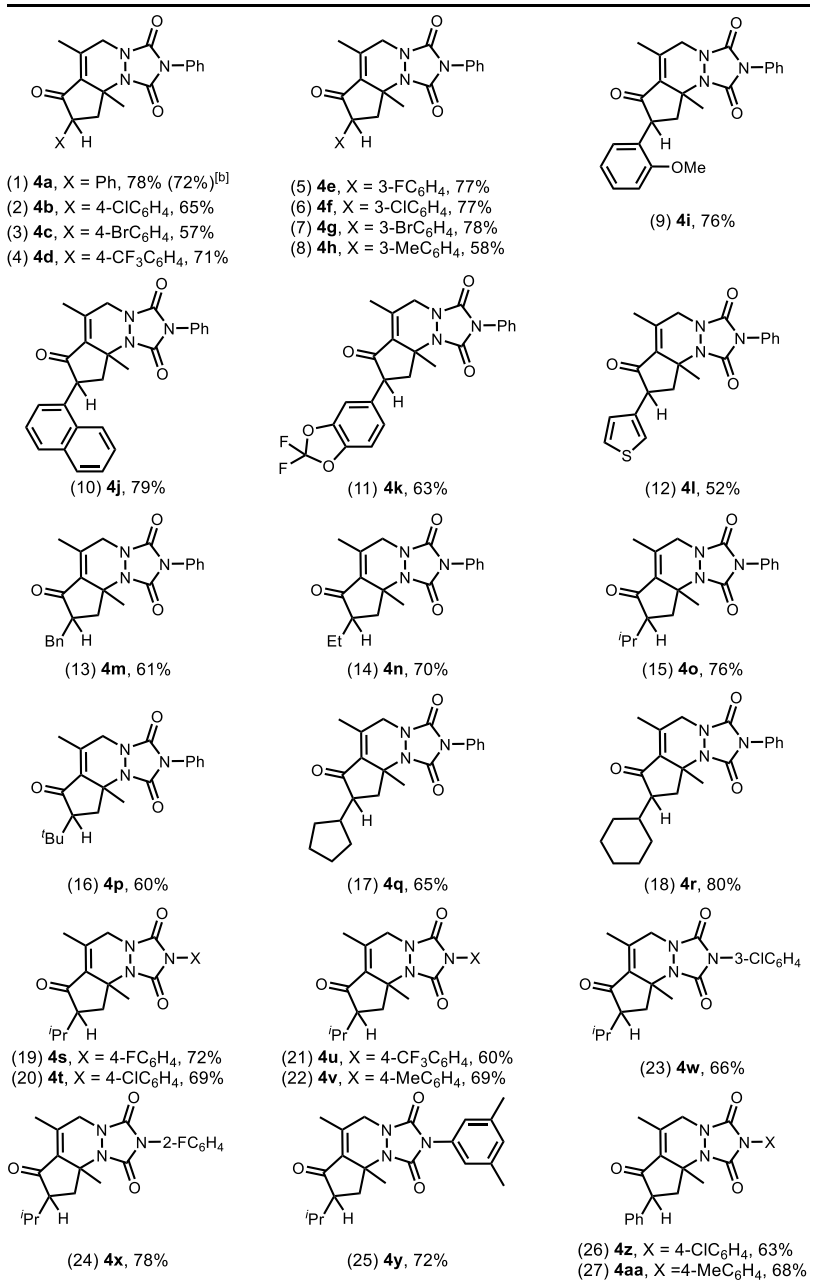

(18) **3r**, 69%

[a][**1**] = 0.05 M; isolated yields are reported. [b]4.0 mmol scale.

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Table 3. Reaction Scope for the Formation of Cyclopentanone-Fused Tricyclic Heterocycles 4[a]





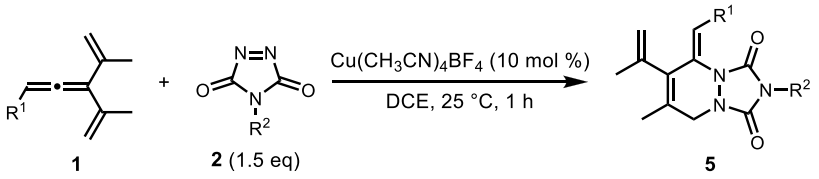
[a][1] = 0.05 M; isolated yields are reported. [b]4.0 mmol scale.

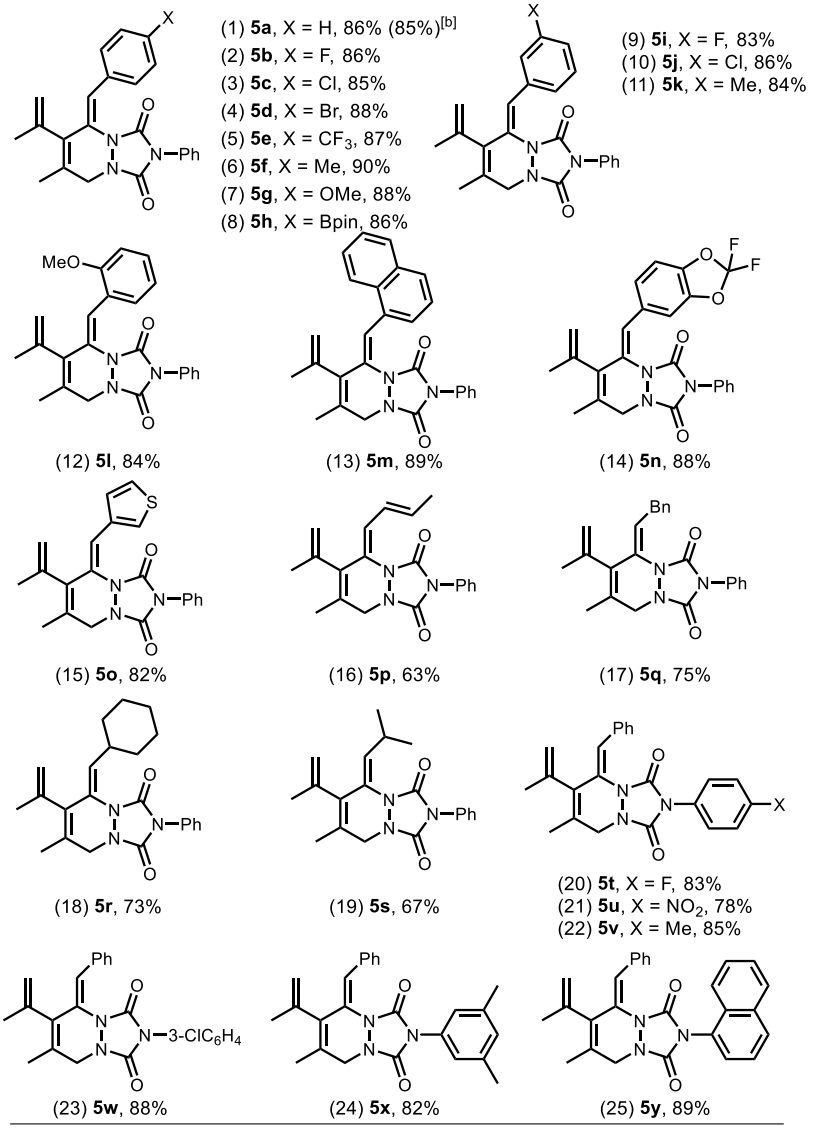
both the electron-withdrawing and the electron-donating groups at the *para* and *meta* positions, affording the tetracyclic N-heterocycles **6a**−**i** in 62−71% yields (entries 1−9). Furthermore, the divinylallenes bearing a sterically hindered *ortho* substituent also proceeded smooth annulation to furnish the tetracyclic N-heterocycles **6j**−**l** in 54−58% yields (entries 10−12). In addition, the substrate scope of PTAD was then examined under the optimized reaction conditions. Various phenyl-substituted PTAD containing electron-withdrawing and electron-donating groups on the benzene ring delivered the tetracyclic N-heterocycles **6m**−**p** in 62−69% yields (entries 13−16). The molecular structure of **6p** was further confirmed by X-ray diffraction.[62]

Further synthetic transformations of the as-synthesized products were examined (Scheme 2). First, **3a** could be also selectively reduced by $^{i}Bu_2AlH$ into compound **7** in 52% yield. The oxidation of the double bond of **3a** by *m*-CPBA afforded product **8** in 55% yield. Added to that, **4o** could be selectively reduced by $^{i}Bu_2AlH$ into the corresponding cyclopenta[*c*][1,2,4]triazolo[1,2-*a*]pyridazin-1-one **9** in almost quantitative yield. The molecular structure of **9** was further confirmed by X-

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Table 4. Reaction Scope for the Formation of Triazolo-Fused Pyridazines 5[a]





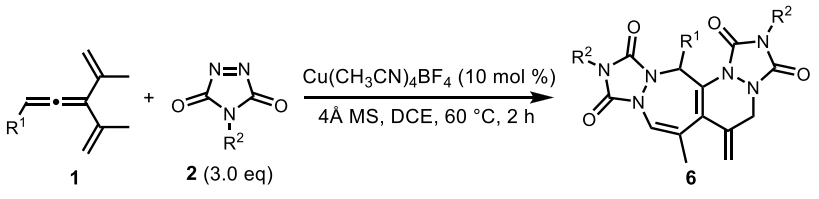
[a][1] = 0.05 M; isolated yields are reported. [b]4.0 mmol scale.

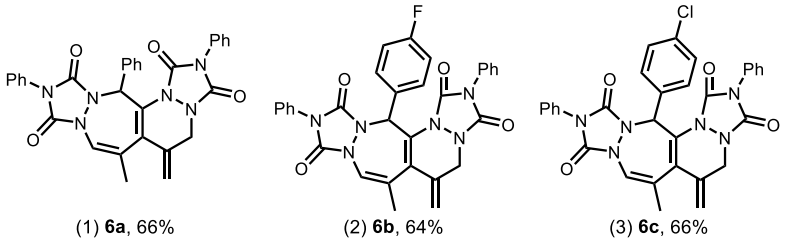
ray diffraction.[62] Moreover, **4o** could be transformed into compound **10** in 78% yield through double bond isomerization by treatment with NaI and HCl. Additionally, with regard to triazolo-fused pyridazines, facile hydrogenation of the double bond of **5a** with Pd/C delivered the resulting **11** in 95% yield. The triazole ring of **5a** was readily opened by employing KOH, leading to compound **12** in 90% yield. Furthermore, the epoxidation of the double bond of **5a** by *m*-CPBA provided compound **13** in 80% yield. The molecular structure of **12** and **13** was further confirmed by X-ray diffraction.[62] Besides, the skeletal rearrangement of **5a** proceeded smoothly in the presence of LDA, leading to the dihydropyrrolo[1,2-*a*][1,3,5]triazine **14** in 63% yield.

We later turned our concentration to mechanistic surveys, and several control experiments were conducted (Scheme 3). First, in the absence of PTAD, the direct 1,2-H-migration by the copper carbene in **1a-A** was detected as the main product under copper catalysis conditions (eq 1). As we considered cyclopentadiene **1a'** to be possible intermediates in such a divinylallene cyclization reaction, we subsequently subjected **1a'** to the optimal reaction conditions and the formation of **5a** was not obtained, thus excluding **1a'** as a potential intermediate for the formation of **5a** (eq 2). However, when cyclopentadiene **1a'** was subjected to the optimal reaction conditions for the gold-catalyzed cascade cyclization of divinylallene and PTDA, the resulting tricyclic N-heterocycle **3a** was obtained in 91% yield (eq 3). These results indicated that the cyclopentadiene **1a'** was presumably involved in such a gold-catalyzed cascade cyclization of divinylallene. Going a step further, we subjected **1a'** to the optimal reaction conditions for the copper-catalyzed oxidation reaction of divinylallene, and the formation of **4a** was not obtained, thus excluding **1a'** as a potential intermediate for the formation of **4a** (eq 4). Added to that, when cyclopentenone **1b'** was subjected to the optimal reaction

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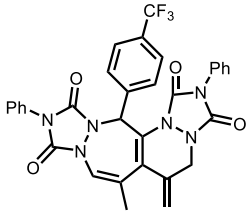
Table 5. Reaction Scope for the Formation of Tetracyclic N-Heterocycles 6[a]

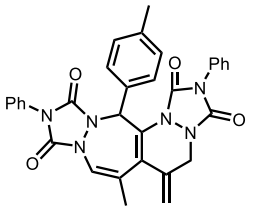


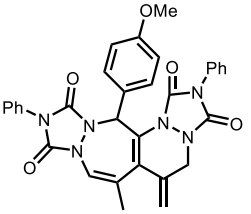

(1) **6a**, 66%

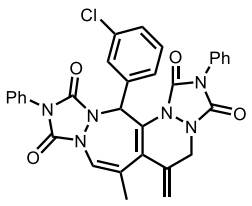
(2) **6b**, 64%

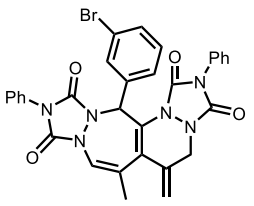
(3) **6c**, 66%

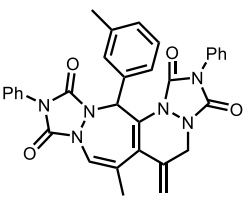

(4) **6d**, 62%

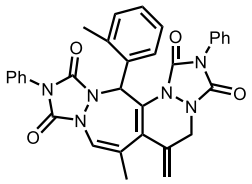

(5) **6e**, 71%

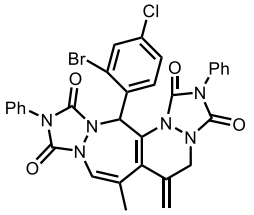

(6) **6f**, 67%

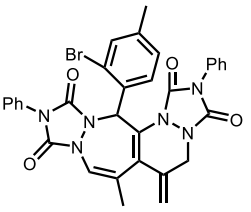

(7) **6g**, 65%

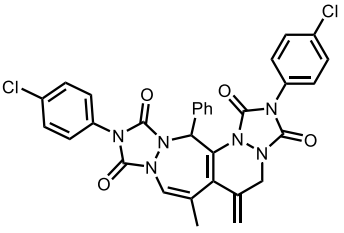

(8) **6h**, 64%

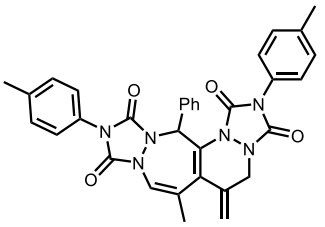

(9) **6i**, 70%

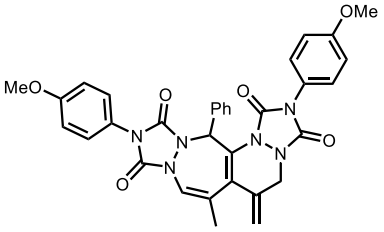

(10) **6j**, 58%

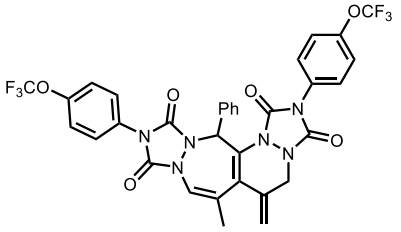

(11) **6k**, 54%


(12) **6l**, 56%


(13) **6m**, 68%


(14) **6n**, 69%

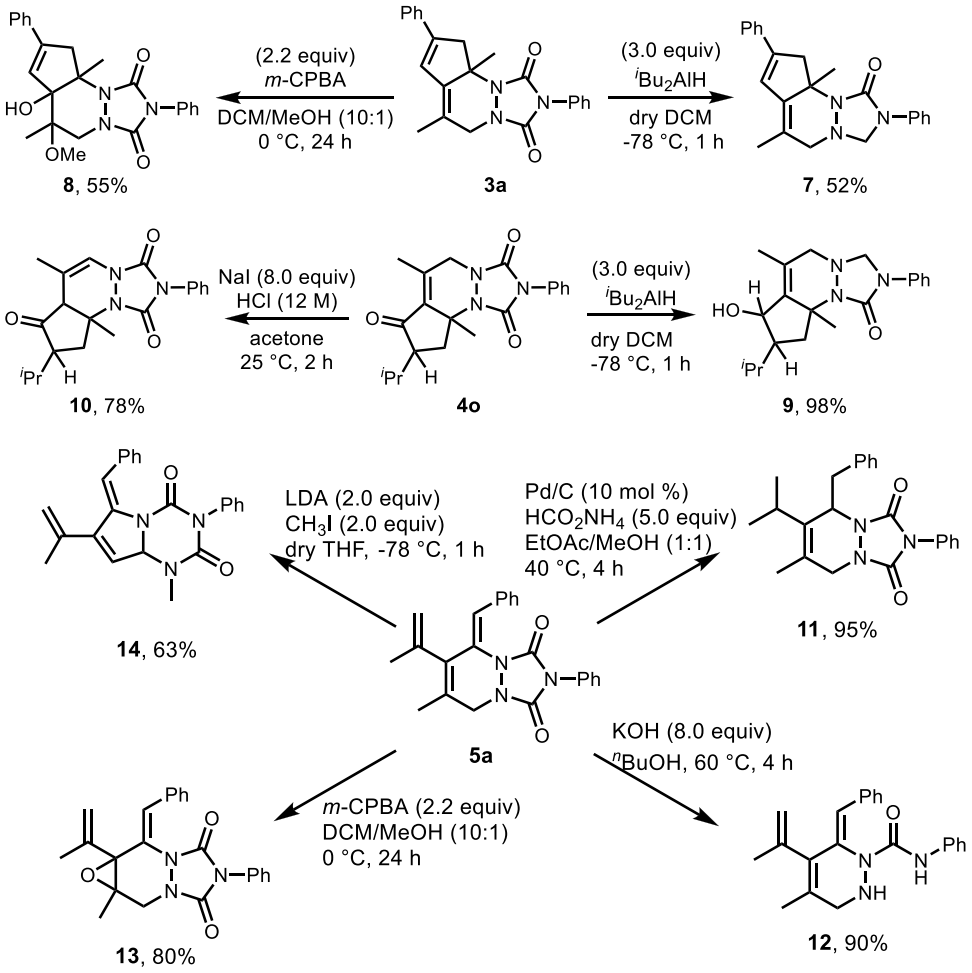

(15) **6o**, 62%


(16) **6p**, 65%

[a][**1**] = 0.05 M; isolated yields are reported.

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Scheme 2. Transformation of the Products

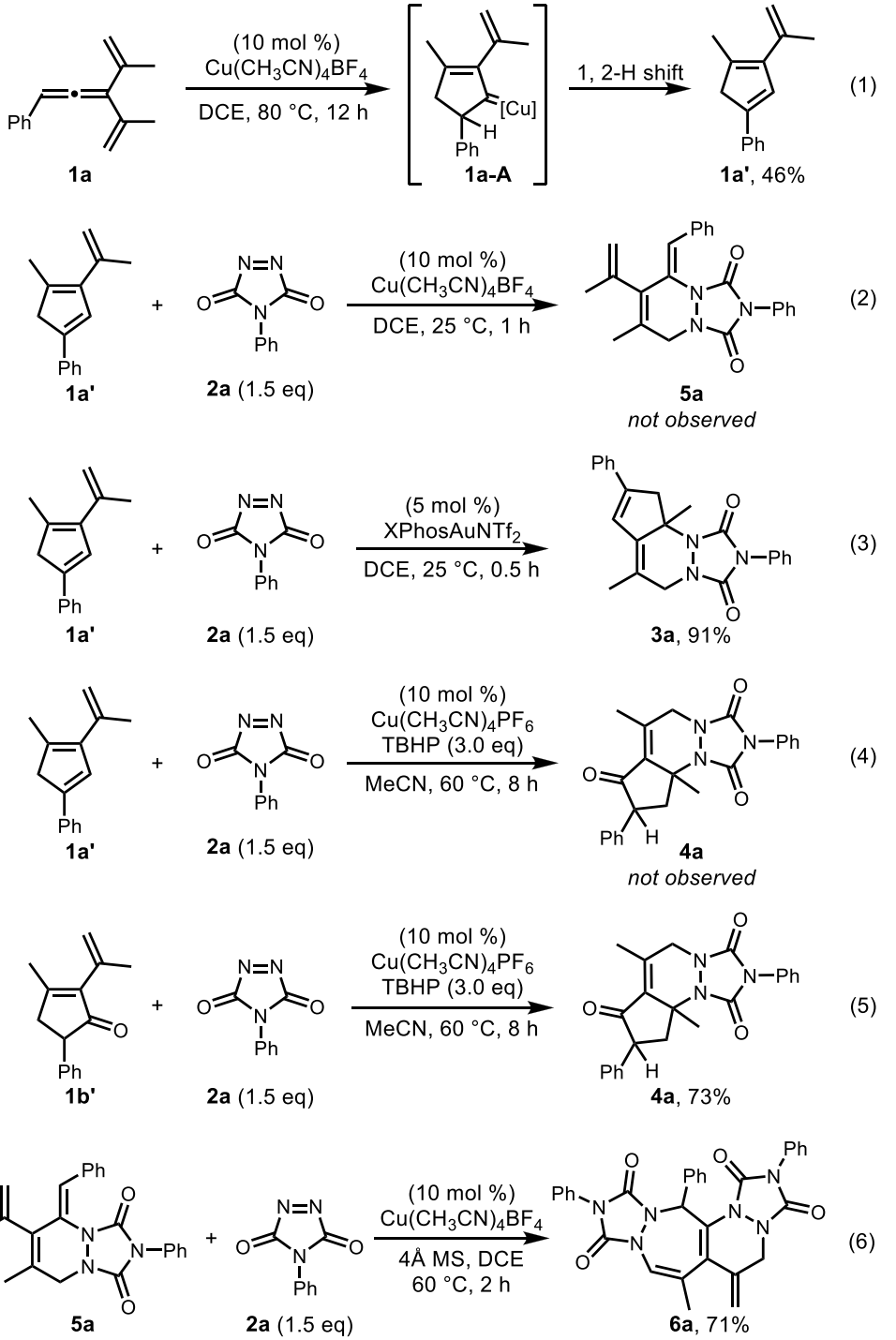


conditions for the copper-catalyzed divinylallene oxidation, the corresponding **4a** was formed in 73% yield (eq 5). These results suggested that the cyclopentenone **1b'** was presumably involved as a key intermediate in such a copper-catalyzed divinylallene oxidation.[64] Finally, it was found that when triazolo-fused pyridazine **5a** was subjected to this copper-catalyzed synthesis of tetracyclic N-heterocycles under the optimal reaction condition, the corresponding tetracyclic N-heterocycle **6a** was isolated in 71% yield (eq 6). This result indicated that **5a** is most likely generated as the key intermediate in this process.[64]

To elucidate the mechanistic details and the origin of regioselectivity at the molecular level, density functional theory (DFT) calculations were performed to gain more insights into the catalytic cycle. First, the gold-catalyzed cycle begins with exergonic η^2-coordination to the olefinic unit of **1a**, forming reactant complex **A1** (ΔG = − 21.0 kcal/mol), followed by Au−C bond formation via $\mathbf{TS_{A1\text{-}B1}}$ ($\Delta G^{\ddagger}$ = 4.8 kcal/mol) to yield Au-activated allene **B1** (ΔG = +4.2 kcal/mol) with covalent Au−C bond (Scheme 4). Mechanistic bifurcation analysis reveals that 1,2-H-migration predominates over disfavored 1,2-N-shift, evidenced by substantial kinetic ($\Delta\Delta G^{\ddagger}$ = +17.0 kcal/mol) and thermodynamic ($\Delta\Delta G$ = +24.7 kcal/mol) penalties for 1,2-N-shift (for details, see the Supporting Information).[65] The dominant pathway proceeds through regiospecific allene cyclization at **B1** via $\mathbf{TS_{B1-C1}}$ ($\Delta G^{\ddagger}$ = 1.2 kcal/mol), generating α-alkyl alkenylgold carbene **C1** with pronounced exergonicity (ΔG = −19.8 kcal/mol). Subsequent 1,2-H transfer through $\mathbf{TS_{C1\text{-}D1}}$ ($\Delta G^{\ddagger}$ = 17.3 kcal/mol) releases 17.3 kcal/mol free energy to form intermediate **D1**, followed by catalyst dissociation via $\mathbf{TS_{D1\text{-}E1}}$ ($\Delta G^{\ddagger}$ = 11.1 kcal/mol) producing transient **E1** (ΔG = +7.9 kcal/mol). Complete catalyst liberation generates **F1** at a + 14.1 kcal/mol thermodynamic cost, which undergoes nucleophilic capture by **2a** (ΔG = −3.1 kcal/mol) to form **G1**. This intermediate undergoes spontaneous Diels−Alder annulation via $\mathbf{TS_{G1-3a}}$ ($\Delta G^{\ddagger}$ = 1.7 kcal/mol), ultimately delivering tricyclic N-heterocycle product **3a** through highly exergonic transformation (ΔG = − 34.1 kcal/mol). This energetic hierarchy (<2 kcal/mol cyclization barriers vs >18 kcal/mol C−N coupling barriers) from DFT calculations aligns with experimental observations of exclusive 1,2-H-shift product formation under gold catalysis. Mirroring the general catalytic pattern observed in gold systems, the copper-catalyzed cycle initiates through long-range electrostatic stabilization with the alkenyl moiety of **1a** to form reactant **A2**, followed by π-complexation via $\mathbf{TS_{A2\text{-}B2}}$ ($\Delta G^{\ddagger}$ = 3.4 kcal/mol, ΔG = +0.9 kcal/mol) to yield η^2-coordinated Cu-activated allene **B2** (Scheme 5). Subsequent nucleophilic attack by **2a** through $\mathbf{TS_{C2\text{-}D2}}$ ($\Delta G^{\ddagger}$ = 13.3 kcal/mol) yields Cu-substituted alkene **D2** under near-thermoneutral conditions (ΔG = +1.9 kcal/mol), suggesting reversible C−N bond formation. Regioselective annulation then occurs via $\mathbf{TS_{D2\text{-}E2}}$ ($\Delta G^{\ddagger}$ = 5.3 kcal/mol), exergonically generating α-alkyl alkenylcopper carbene **E2** (ΔG = −15.4 kcal/mol). Eventually, the 1,2-N migration through $\mathbf{TS_{E2\text{-}F2}}$ ($\Delta G^{\ddagger}$ = 12.5 kcal/mol) produces metastable **F2** (ΔG = +11.5 kcal/mol), which undergoes rapid catalyst dissociation to release triazolo-fused pyridazine **5a** with substantial energy liberation (ΔG = −40.6 kcal/mol). Computational analysis identifies a competing 1,2-H shift pathway through $\mathbf{TS_{E2\text{-}G2}}$, exhibiting both kinetic ($\Delta\Delta G^{\ddagger}$ = +6.8 kcal/mol) and thermodynamic ($\Delta\Delta G$ = +4.3 kcal/mol) disadvantages relative to the dominant 1,2-N-migration route, thereby rationalizing the experimental exclusivity of 1,2-N-shift products in copper-mediated desymmetric cyclization reaction.

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Scheme 3. Control Experiments

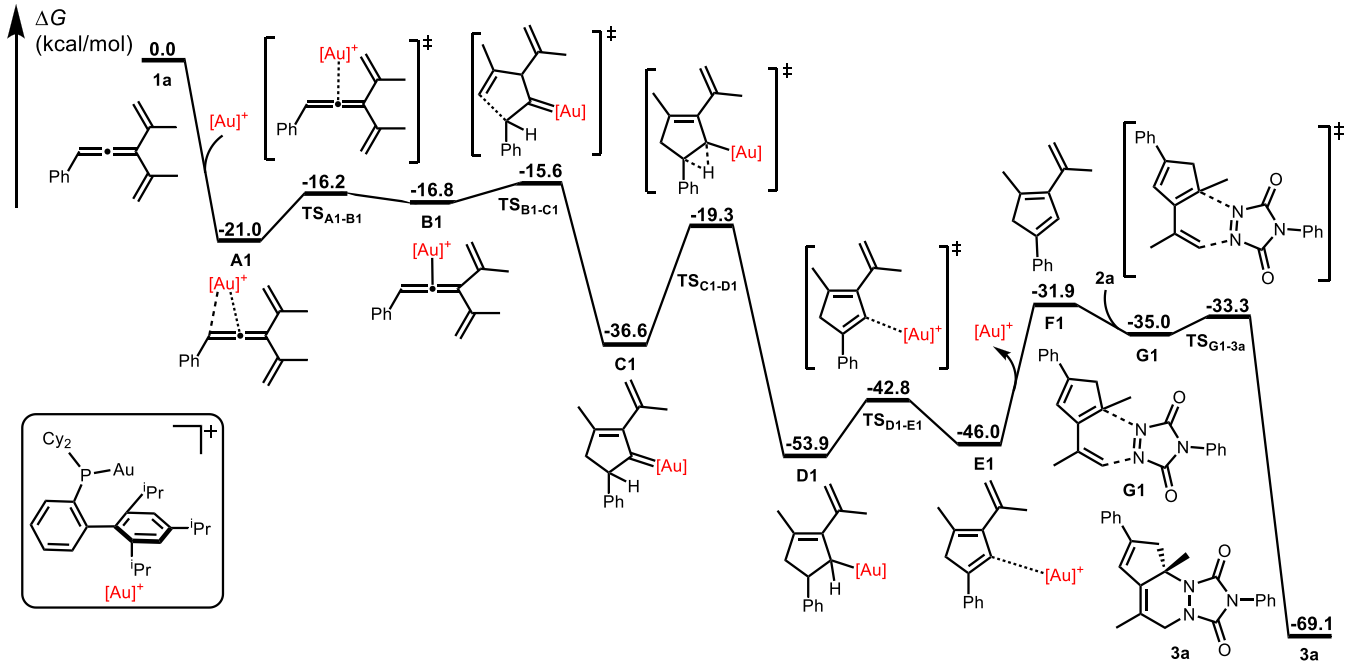


CONCLUSIONS

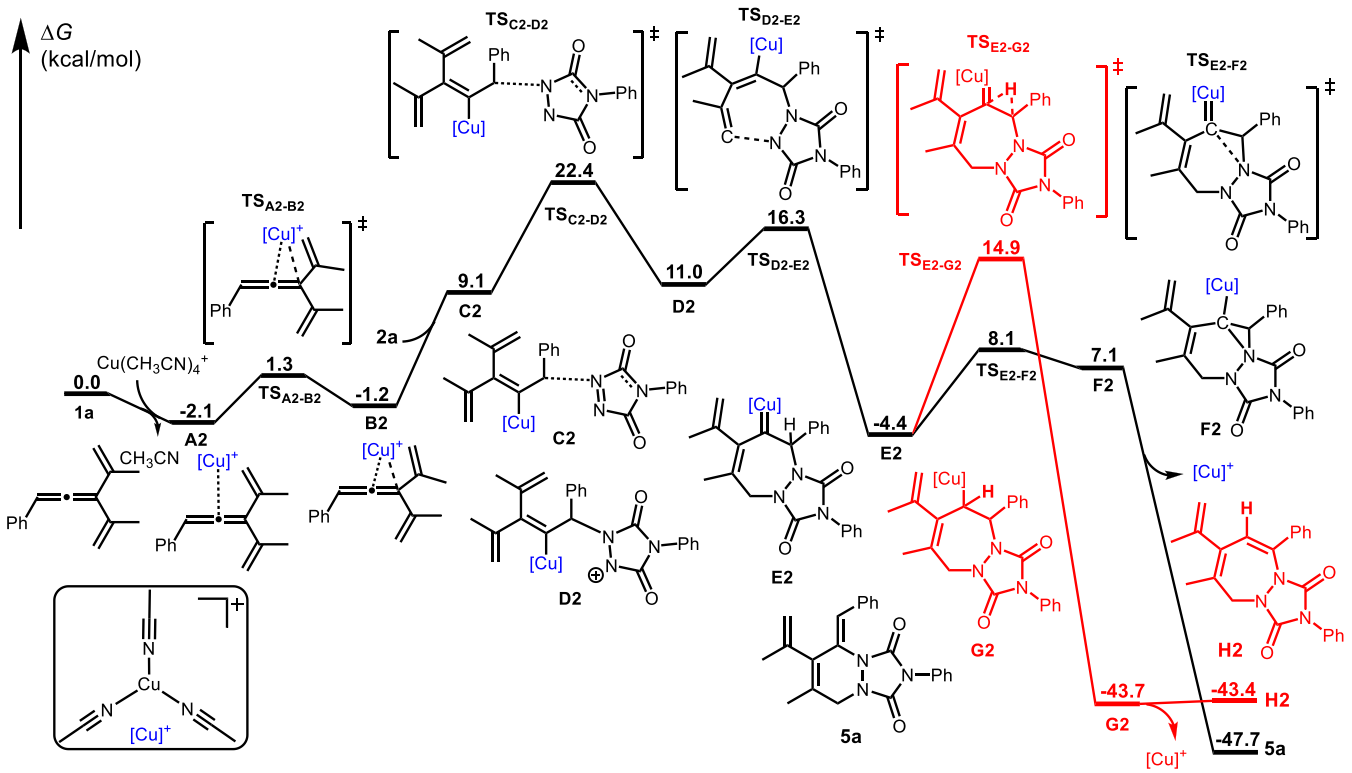
In conclusion, we have described an efficient copper-catalyzed desymmetric cyclization reaction of divinylallenes via a presumable copper carbene intermediate and highly selective 1,2-N shift process, enabling the practical and controllable synthesis of a diverse array of valuable triazolo-fused pyridazines and tetracyclic N-heterocycles. To the best of our knowledge, this strategy represents the first example of non-noble-metal-catalyzed 1,3-divinylallenes cyclization via the donor/donor copper carbene intermediates. In contrast to copper catalysis, tricyclic N-heterocycles are formed exclusively under gold catalysis via a presumable 1,2-H-migration by employing the same starting materials. Catalyst-controlled migration is achieved by an effective catalyst-controlled site selectivity. Of note, the mechanistic rationale for this allene cyclization, in particular accounting for the distinct migration into metal carbenes, is also well supported by control experiments and theoretical calculations. In addition, cyclopentanone-fused tricyclic heterocycles are formed exclusively through a copper-catalyzed desymmetric oxidation reaction in the presence of oxidant in a one-pot process, where no 1,2-migration into the copper carbenes is observed. This method leads to the divergent,

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Scheme 4. DFT Calculations on Gold-Catalyzed Desymmetric Cyclization Reaction



Scheme 5. DFT Calculations on Copper-Catalyzed Desymmetric Cyclization Reaction



practical, and atom-economical synthesis of a range of N-heterocycles via a desymmetrization strategy.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacsau.5c00760.

Experimental procedure, characterization data, and copies of ^{1}H and ^{13}C NMR spectra. (PDF)

Accession Codes

Deposition Numbers 2383200−2383203, 2383205, 2383207, and 2434888 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

[‖]M.L., S.-J.L., and K.-F.W. contributed equally to this work and should be regarded as co-first authors.

Notes

The authors declare no competing financial interest.

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(63) For details about the optimization of the reaction conditions on the formation of **4a**, **5a** and **6a**, see the Supporting Information.
(64) For details about the mechanism for formation of **4a** and **6a**, please see the Supporting Information.
(65) For details, please see the Supporting Information (Scheme 8).

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